

Convergencia fuerte implica débil

Proposición 1. Sea X un espacio normado, $(x_n)_{n \in \mathbb{N}} \subset X$ y $x \in X$. Entonces,

$$x_n \rightarrow x \implies x_n \rightharpoonup x.$$

Demostración: Como $L \in X^*$ es continua, $\|x_n - x\| \rightarrow 0 \implies L(x_n) \rightarrow L(x)$. ■

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